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Lecture 06A
Before-Tutorial-Session
Tutorial 5

https://app.wooclap.com/ **HIDDEN**

Recall:

do main

codomain

Definition. Let $\boldsymbol{T} : \mathcal{V} \to \mathcal{W}$ be a linear map.

Let $\mathcal{B} = \{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\}$ be a basis for **domain $\mathcal{V}$** and $\mathcal{A} = \{\boldsymbol{w}_1, \dots, \boldsymbol{w}_m\}$ be a basis for **codomain $\mathcal{W}$**.

Then the *matrix representation with respect to bases $\mathcal{A}$ and $\mathcal{B}$*, denoted by $[\boldsymbol{T}]_{\mathcal{B}}^{\mathcal{A}}$, is the matrix such that

don't memorise this formula

$$\boldsymbol{T}(\boldsymbol{v}_j) = \underline{\boldsymbol{T}_{1,j}\boldsymbol{w}_1 + \cdots + \boldsymbol{T}_{m,j}\boldsymbol{w}_m}.$$

Instead, note: · size of matrix

- how to index the rows and columns.

$$[\boldsymbol{T}]_{\mathcal{B}}^{\mathcal{A}}$$

$\mathcal{A}$: basis for codomain

$\mathcal{B}$: basis for domain

codomain basis $\mathcal{A}$

$$\begin{array}{c|cccc}
\boldsymbol{w}_1 & \boldsymbol{T}_{1,1} & \boldsymbol{T}_{1,2} & \cdots & \boldsymbol{T}_{1,n} \\
\boldsymbol{w}_2 & \boldsymbol{T}_{2,1} & \boldsymbol{T}_{2,2} & \cdots & \boldsymbol{T}_{2,n} \\
\vdots & \vdots & \vdots & \ddots & \vdots \\
\boldsymbol{w}_m & \boldsymbol{T}_{m,1} & \boldsymbol{T}_{m,2} & \cdots & \boldsymbol{T}_{m,n} \\
\hline
 & \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n
\end{array}$$

domain basis $\mathcal{B}$

Question 8. Define $T : \mathcal{M}_{2\times 2}(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ by

polynomial

matrix

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b) + 2dx + bx^2.$$

Let

$$\mathcal{B} = \left\{E_{1,1} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, E_{1,2} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, E_{2,1} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, E_{2,2} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right\} \text{ and } \mathcal{A} = \{1, x, x^2\}$$

What is $[T]_{\mathcal{B}}^{\mathcal{A}}$?

$\{1, x, x^2\}$ codomain basis $\left\{ \underbrace{[\hat{T}]}_{\text{domain}} \right._{\mathcal{B}}^{\mathcal{A}}$

$\{\hat{E}_{11}, \hat{E}_{12}, \hat{E}_{21}, \hat{E}_{22}\}$

1. A 4×3-matrix whose entries belong to $\mathbb{R}$.
2. A 3×4-matrix whose entries belong to $\mathbb{R}$. ✓
3. A 4×3-matrix whose entries belong to $\mathcal{P}_2(\mathbb{R})$.
4. A 3×4-matrix whose entries belong to $\mathcal{P}_2(\mathbb{R})$.

Question 8. Define $T : \mathcal{M}_{2\times 2}(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ by

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b) + 2dx + bx^2.$$

Let

$$\mathcal{B} = \left\{ E_{1,1} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, E_{1,2} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, E_{2,1} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, E_{2,2} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} \text{ and } \mathcal{A} = \{1, x, x^2\}$$

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$$[T]_{\mathcal{B}}^{\mathcal{A}}$$

$T(E_{11}) = 1$
$= 1 + 0x + 0x^2$

1	1	1	0	0
x	0	0	0	2
x^2	0	1	0	0
	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

What is the column corresponding to $E_{1,1}$? $(1,0,0)^T$

Question 8. Define $T : \mathcal{M}_{2\times 2}(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ by

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b) + 2dx + bx^2.$$

Let

$$\mathcal{B} = \left\{ E_{1,1} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, E_{1,2} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, E_{2,1} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, E_{2,2} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} \text{ and } \mathcal{A} = \{1, x, x^2\}$$

$$[T]_{\mathcal{B}}^{\mathcal{A}}$$

$$\begin{array}{c|c}
1 & \\
x & \text{HIDDEN} \\
x^2 & \\
\hline
 & \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}
\end{array}$$

What is the column corresponding to $E_{1,2}$? $(1,0,1)^T$

Question 8. Define $T : \mathcal{M}_{2\times 2}(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ by

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b) + 2dx + bx^2.$$

Let

$$\mathcal{B} = \left\{ E_{1,1} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, E_{1,2} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, E_{2,1} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, E_{2,2} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} \text{ and } \mathcal{A} = \{1, x, x^2\}$$

$$[T]_{\mathcal{B}}^{\mathcal{A}}$$

$$T\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = 0$$

$$T\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = 2x$$

1

x

x^2

HIDDEN

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

What is $[T]_{\mathcal{B}}^{\mathcal{A}}$? $\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \end{bmatrix}$

Question 8. Define $T : \mathcal{M}_{2\times 2}(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ by

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b) + 2dx + bx^2.$$

Let

$$\mathcal{B} = \left\{ E_{1,1} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, E_{1,2} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, E_{2,1} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, E_{2,2} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} \text{ and } \boxed{\mathcal{C} = \{x^2, x^2+1, x^2+x\}}$$

diff basis from usual

usually is standard basis

$$T\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = 1$$

changed!

	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$
x^2				
x^2+1				
x^2+x				

Is $(1,0,0)^T$ the column corresponding to $E_{1,1}$? No

$$[T]_{\mathcal{B}}^{\mathcal{C}}$$

Question 8. Define $T : \mathcal{M}_{2\times 2}(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ by

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b) + 2dx + bx^2.$$

Let

$$\mathcal{B} = \left\{ E_{1,1} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, E_{1,2} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, E_{2,1} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, E_{2,2} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} \text{ and } \boxed{\mathcal{C} = \{x^2, x^2+1, x^2+x\}}$$

$$T\begin{bmatrix} \cancel{0}^{\,1} & 0 \\ 0 & 0 \end{bmatrix} = 1$$

Solve $1 = \lambda_1(x^2) + \lambda_2(1+x^2) + \lambda_3(x+x^2)$

3 equations in 3 unknowns.

$-1x^2$

$1 + 1x^2$

$0x + 0x^2$

$\lambda_1 = -1$

$\lambda_2 = 1$

$\lambda_3 = 0$

	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$
x^2	λ_1			
$1 + x^2$	λ_2			
$x + x^2$	λ_3			

$$[T]_{\mathcal{B}}^{\mathcal{C}}$$

What is the column corresponding to $E_{1,1}$? $(-1, 1, 0)^T$

Question 8. Define $T : \mathcal{M}_{2\times 2}(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ by

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b) + 2dx + bx^2.$$

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Let

$$\mathcal{B} = \left\{ E_{1,1} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, E_{1,2} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, E_{2,1} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, E_{2,2} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} \text{ and } \boxed{\mathcal{C} = \{x^2, x^2+1, x^2+x\}}$$

$$T\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = 1 + x^2$$

$$= \lambda_1(x^2) + \lambda_2(1+x^2) + \lambda_3(x+x^2)$$

$\lambda_1 = 0$

$\lambda_2 = 1$

$\lambda_3 = 0$

	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$
x^2	HIDDEN			
$1+x^2$				
$x+x^2$				

$$[T]_{\mathcal{B}}^{\mathcal{C}}$$

What is the column corresponding to $E_{1,2}$?

Question 8. Define $T : \mathcal{M}_{2\times 2}(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ by

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b) + 2dx + bx^2.$$

Let

$$\mathcal{B} = \left\{ E_{1,1} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, E_{1,2} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, E_{2,1} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, E_{2,2} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} \text{ and } \boxed{\mathcal{C} = \{x^2, x^2+1, x^2+x\}}$$

$$[T]_{\mathcal{B}}^{\mathcal{C}} \qquad \begin{array}{cc} x^2 & \\ x^2+1 & \boxed{\text{HIDDEN}} \\ x^2+x & \\ & \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \end{array}$$

DIY

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Question 8. Define $T : \mathcal{M}_{2\times 2}(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ by

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b) + 2dx + bx^2.$$

(c) Using the matrix representations $[T]_{\mathcal{B}}^{\mathcal{A}}$ or $[T]_{\mathcal{B}}^{\mathcal{C}}$, find a basis for $\operatorname{null}(T)$.

$$[T]_{\mathcal{B}}^{\mathcal{A}} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \end{bmatrix} \xrightarrow{\text{rref}} \begin{array}{cccc} a & b & t & d \\ \hline \end{array}\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$\begin{array}{cccc} \Delta & \Delta & \text{not pivot} & \Delta \end{array}$$

Δ: pivot

What is the nullspace of $[T]_{\mathcal{B}}^{\mathcal{A}}$?

free variable : t

pivots $\begin{cases} a + 0t = 0 \Rightarrow a = 0 \\ b + 0t = 0 \Rightarrow b = 0 \\ d + 0t = 0 \Rightarrow d = 0 \end{cases}$

$$\text{nullspace of}\left([T]_{\mathcal{B}}^{\mathcal{A}}\right) = \left\{ \begin{bmatrix} a \\ b \\ t \\ d \end{bmatrix} : a = b = d = 0,\ t \in \mathbb{R} \right\}$$

$$= \operatorname{span} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

Question 8. Define $T : \mathcal{M}_{2\times 2}(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ by

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b) + 2dx + bx^2.$$

(c) Using the matrix representations $[T]^{\mathcal{A}}_{\mathcal{B}}$ or $[T]^{\mathcal{C}}_{\mathcal{B}}$, find a basis for $\operatorname{null}(T)$.

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What is a basis for the nullspace of T ?

basis for nullspace $[\hat{T}]^{\mathcal{A}}_{\mathcal{B}} = \left[\begin{array}{c|c} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} & \begin{array}{c} E_{11} \\ E_{12} \\ E_{21} \\ E_{22} \end{array} \end{array} \right]$

$\downarrow$

basis for nullspace $\hat{T}$ $= \left[\begin{array}{c} 0 \cdot \hat{E}_{11} + 0 \cdot \hat{E}_{12} \\ + 1 \hat{E}_{21} + 0 \cdot \hat{E}_{22} \end{array} \right]$

$= [\hat{E}_{21}]$

What does $\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ represent in the domain $\mathcal{M}_{2\times 2}$?

Recall from MH1200

$$\boldsymbol{M} = \begin{bmatrix} 1 & 2 & 0 & 10 \\ 0 & 0 & 1 & 20 \\ 0 & 0 & 2 & 40 \end{bmatrix} \overset{\text{rref}}{\Longrightarrow} \begin{bmatrix} 1 & 2 & 0 & 10 \\ 0 & 0 & 1 & 20 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

What is the column space of $\boldsymbol{M}$?

$\text{span}\{(1,0,0)^T, (0,1,2)^T\}$

Question 8. Define $T : \mathcal{M}_{2\times 2}(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ by

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b) + 2dx + bx^2.$$

(d) Find a basis for $\operatorname{range}(T)$.

$$[T]_{\mathcal{B}}^{\mathcal{A}} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \end{bmatrix} \xrightarrow{\text{rref}} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

What is the column space of $[T]_{\mathcal{B}}^{\mathcal{A}}$?

column space = span$\left[\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \right]$

Question 8. Define $T : \mathcal{M}_{2\times 2}(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ by

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b) + 2dx + bx^2.$$

(d) Find a basis for $\operatorname{range}(T)$.

What is a basis for the range of T ?

this does not belong to codomain

$$\text{basis for colspace } [T]_B^A = \left[\begin{array}{c} 1 \\ x \\ x^2 \end{array} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \right]$$

$$\text{basis for range}(T) = [\, 1,\ 1+x^2,\ 2x \,]$$